PORTFOLIO PROJECT · DISK UTILITIES

AEGIS DiskGuard

SSD & HDD maintenance toolkit for Windows

Scan. Repair. Protect.



Juan Paolo Dente · C# · WinForms · Raw disk I/O · 13 tools · built without any install

THE PROBLEM

A failing disk, and no clear next step

The upgrade

A 128 GB laptop drive was almost full. Moving Windows to a bigger SSD meant reinstalling everything, or trusting a clone tool that might stall.

The broken drive

RAW, write protected, missing files, a format that never finishes. The fixes are scattered across many tools, and most of them are confusing.

DiskGuard puts the fixes in one app, in plain words, and says honestly what software can and can’t fix.

WHO IT'S FOR

Made for people who aren't disk experts

The worry

"Will I lose my files?" Disk tools are full of words like MBR, GPT and TRIM, and one wrong click can erase a drive.

The gap

Backup, clone, repair and recovery usually live in separate apps. One clone tool I tried sat at 0% with a 14 hour estimate.

What DiskGuard gives

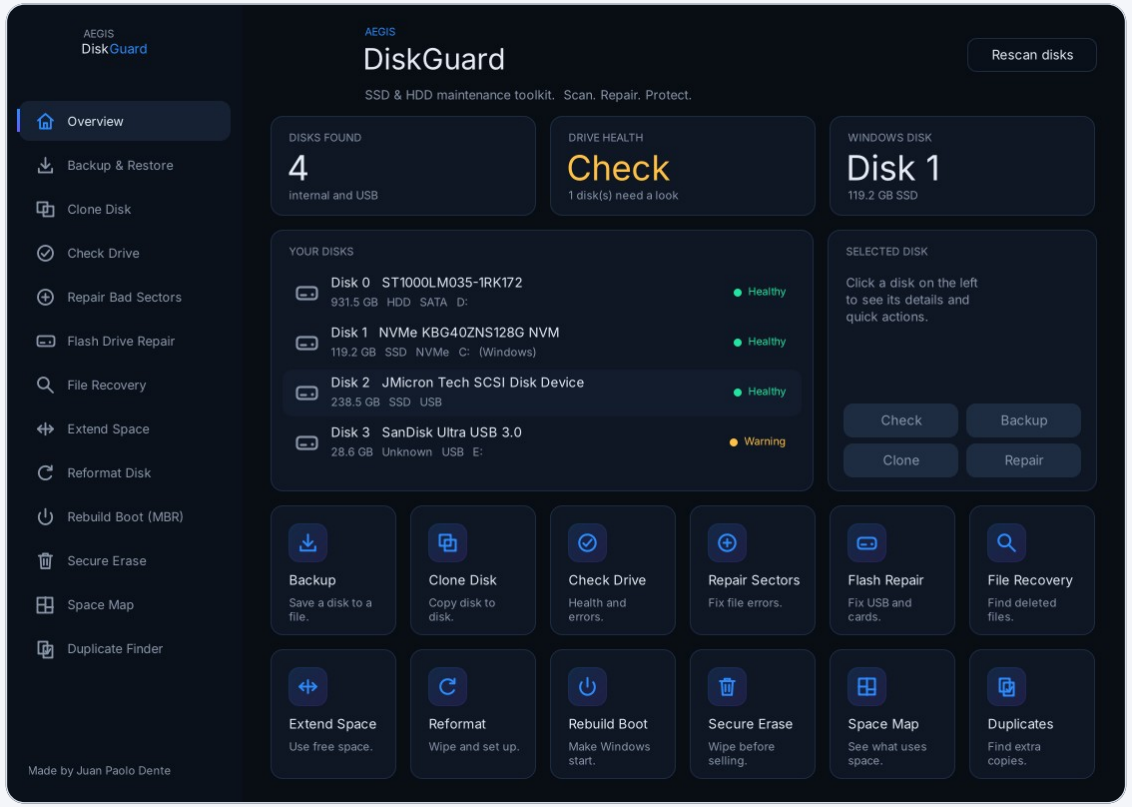
One dark, simple app. Plain words, a warning before anything destructive, and a typed confirmation before a disk is erased.

THE PRODUCT

One app, thirteen tools

A sidebar on the left, a dashboard on the right. Click a disk to see its details, then jump straight into a tool with that disk already chosen.

- Live list of every disk, with health
- Count-up stat cards you can click
- Rescan, then act in one click
- Runs as one .exe, no installer



ARCHITECTURE

Pages on top, Windows tools underneath

Pages and UI

Thirteen pages and a custom dark theme drawn by hand in WinForms. Asks for a disk, shows progress.

Engine

Runs each job on its own thread with cancel, progress and speed. Raw disk reads and writes in 4 MB blocks.

Windows itself

Disk APIs, WMI, diskpart, chkdsk, bcdboot and PowerShell. Nothing extra to install.

Every destructive step goes through one path: check the disk is safe to touch, ask the person to type its number, run, then report in plain words.

THE TOOLKIT · THIRTEEN FUNCTIONS, ONE PER SLIDE NEXT

Scan it, fix it, protect it

Overview
Every disk and its health.

Backup
Disk to one file, and back.

Clone
Copy a whole disk.

Check
Health and errors.

Repair
Bad sectors, TRIM.

Flash
Fix USB sticks.

Recovery
Find lost files.

Reformat
Wipe and set up fresh.

Boot
Rebuild MBR and UEFI.

Erase
Wipe before you sell.

Extend
Grow C: over free space.

Space map
See what uses space.

Duplicates
Remove extra copies.

Twelve are tools that act on a disk. The Overview is the dashboard that starts them.

TOOLKIT · FUNCTION 1 OF 13

Overview dashboard

What it does

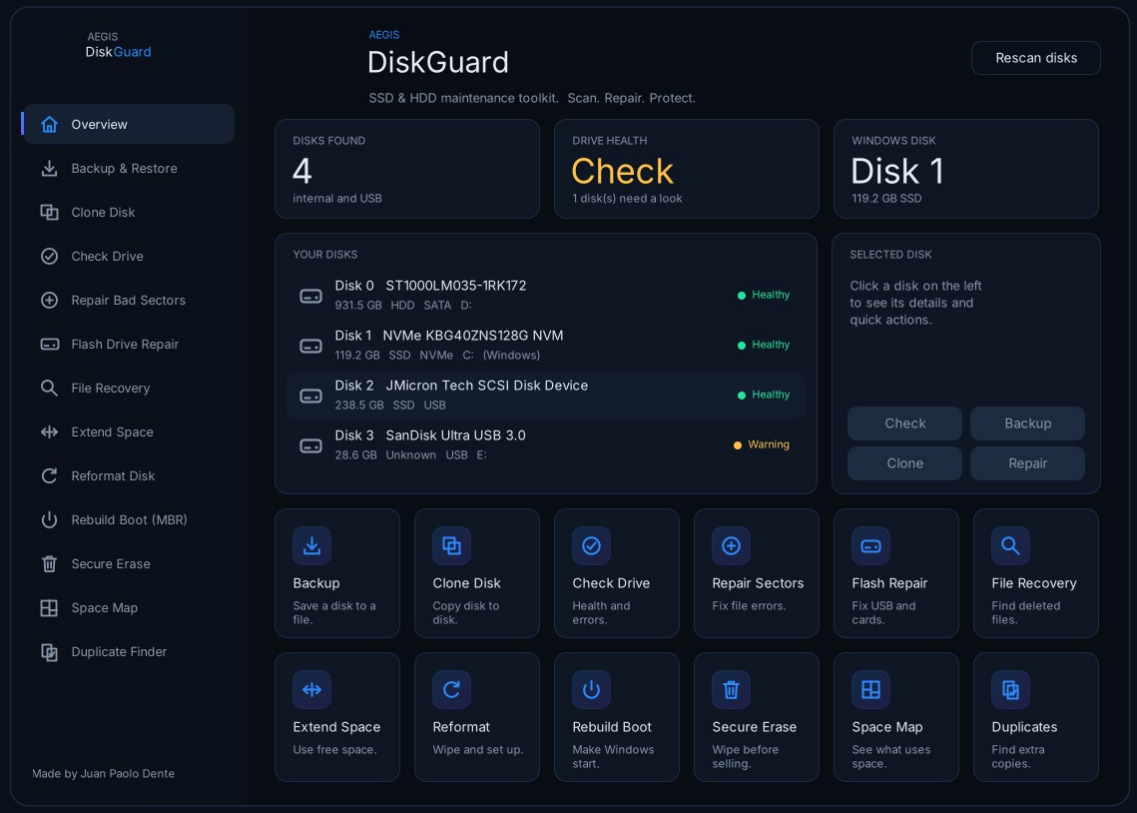
Lists every disk with its size, type, drive letters and health. Three stat cards sum it up.

Why it matters

You see in five seconds if any disk needs you, then act without hunting through menus.

How it works

Reads Windows disk data (WMI). Click a disk for quick actions, a stat card to jump, or Rescan.



TOOLKIT · FUNCTION 2 OF 13

Backup and Restore

What it does

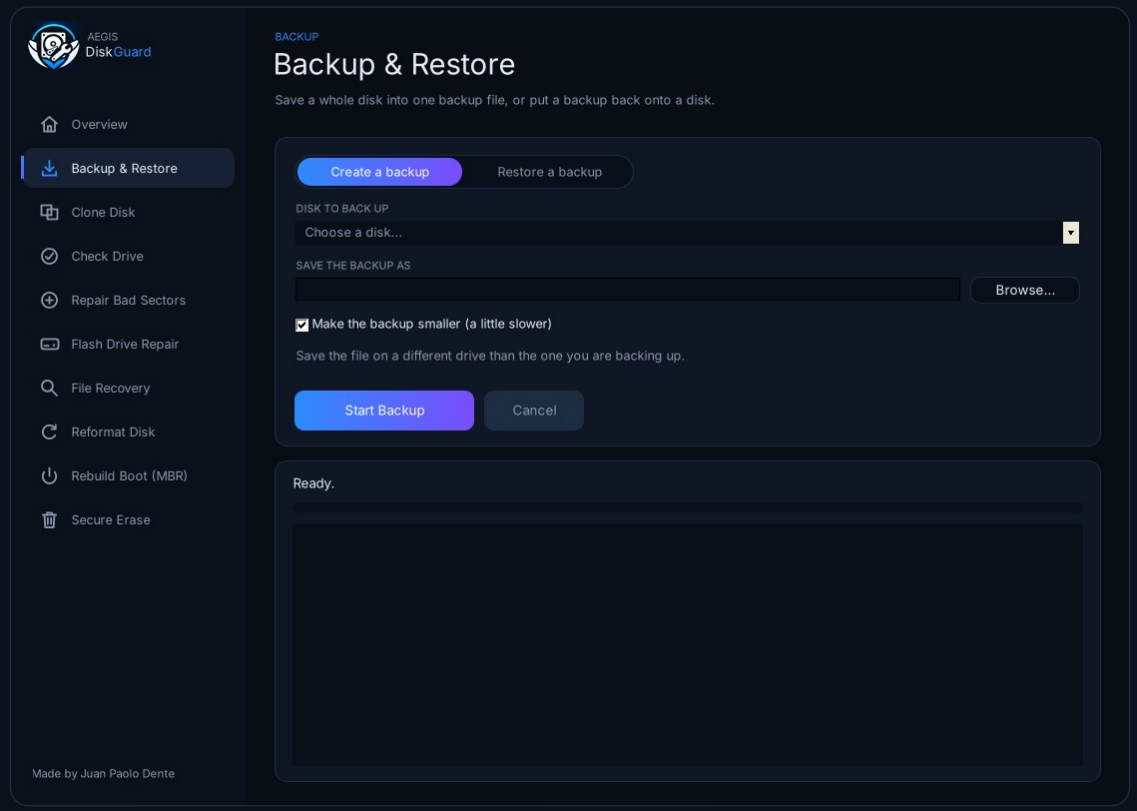
Saves a whole disk into one .edc file, optionally compressed, and puts a backup back onto a disk later.

Why it matters

One file is easy to store, copy and keep safe, and it holds everything, not just chosen folders.

How it works

- Own image format with a small header
- Won’t save the file on the disk being backed up
- Restore asks you to type the disk number



TOOLKIT · FUNCTION 3 OF 13

Clone Disk

What it does

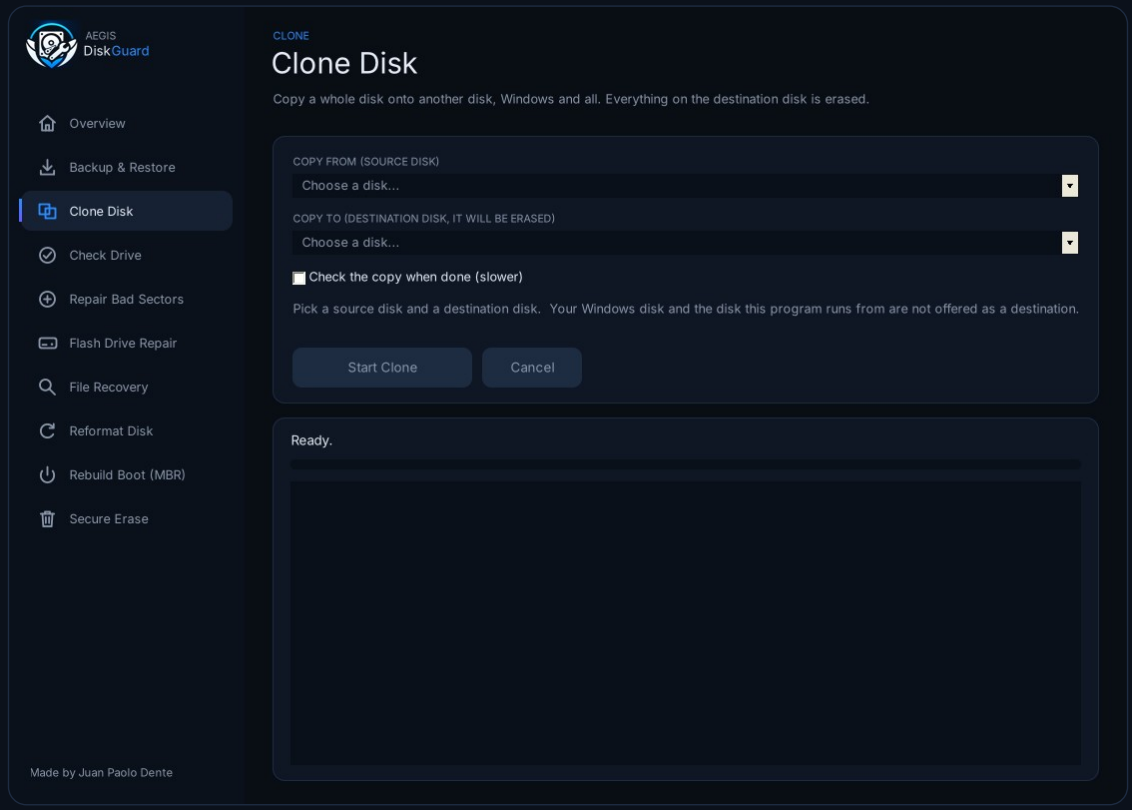
Copies one whole disk onto another, Windows and all, so the new disk can start the computer.

Why it matters

Moving to a bigger SSD without reinstalling Windows or your programs.

How it works

- Raw sector copy, 4 MB at a time
- Optional check of the copy afterwards
- Leaves the clone offline so Windows keeps its disk ID



TOOLKIT · FUNCTION 4 OF 13

Check Drive

What it does

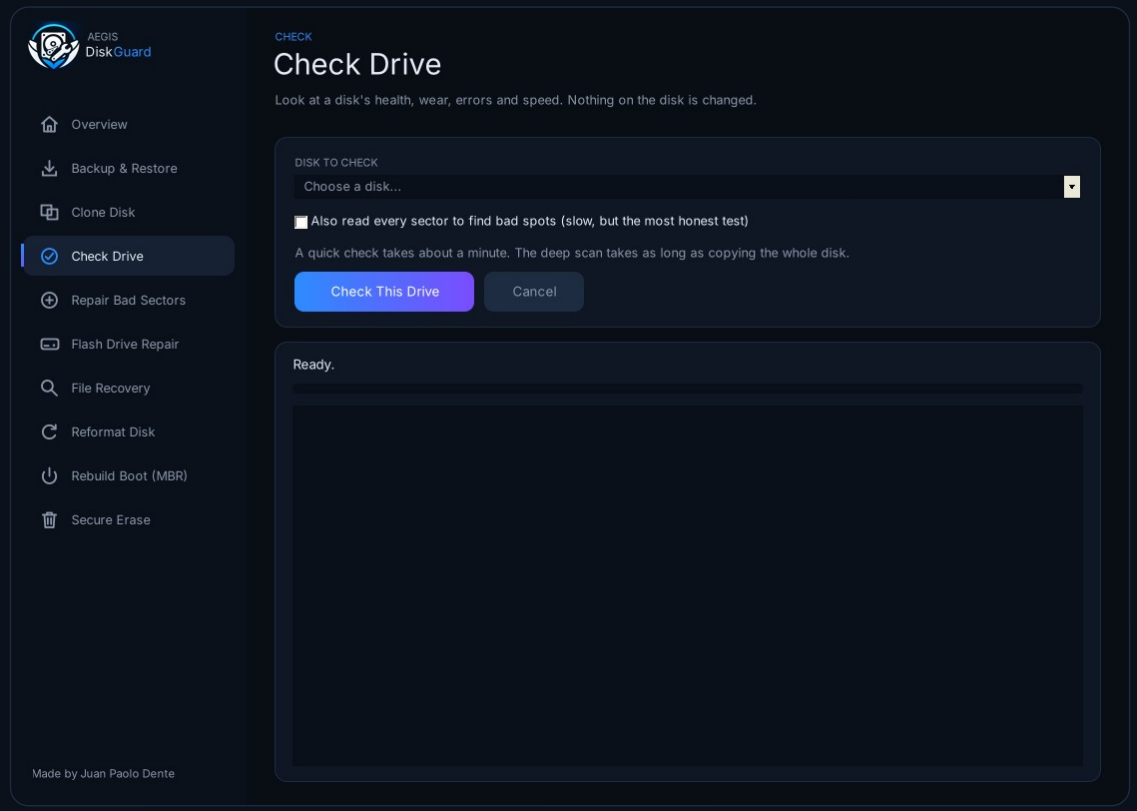
Reads a disk’s health, wear, errors and speed, then gives one verdict: good, needs attention or has problems.

Why it matters

Know a drive is dying before it takes your files with it.

How it works

- Quick check in about a minute
- Optional deep scan reads every sector
- Changes nothing on the disk



TOOLKIT · FUNCTION 5 OF 13

Repair Bad Sectors

What it does

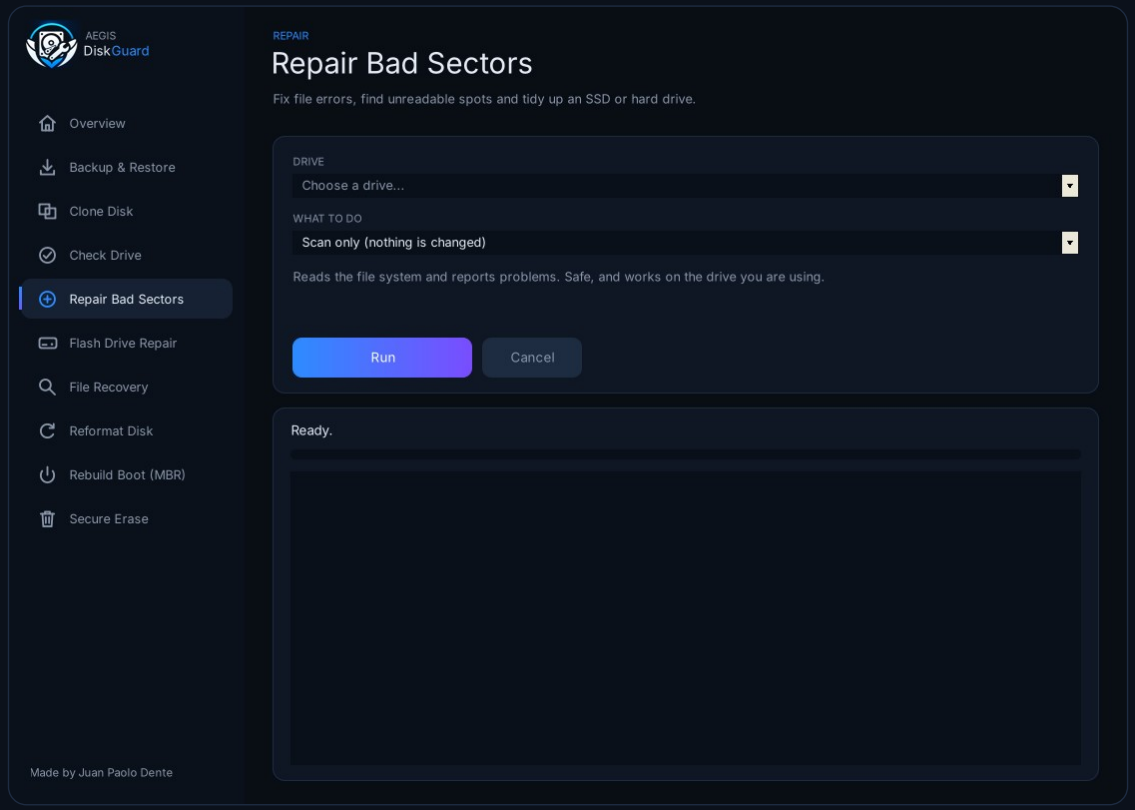
Five modes: scan only, fix file system errors, deep read scan, find and mark bad sectors, and SSD TRIM tune up.

Why it matters

Fixes what software can fix, and says plainly when a disk should be replaced.

How it works

- Uses Windows chkdsk for repairs
- Warns that marking sectors on an SSD does very little
- Deep scan counts unreadable sectors



TOOLKIT · FUNCTION 6 OF 13

Flash Drive Repair

What it does

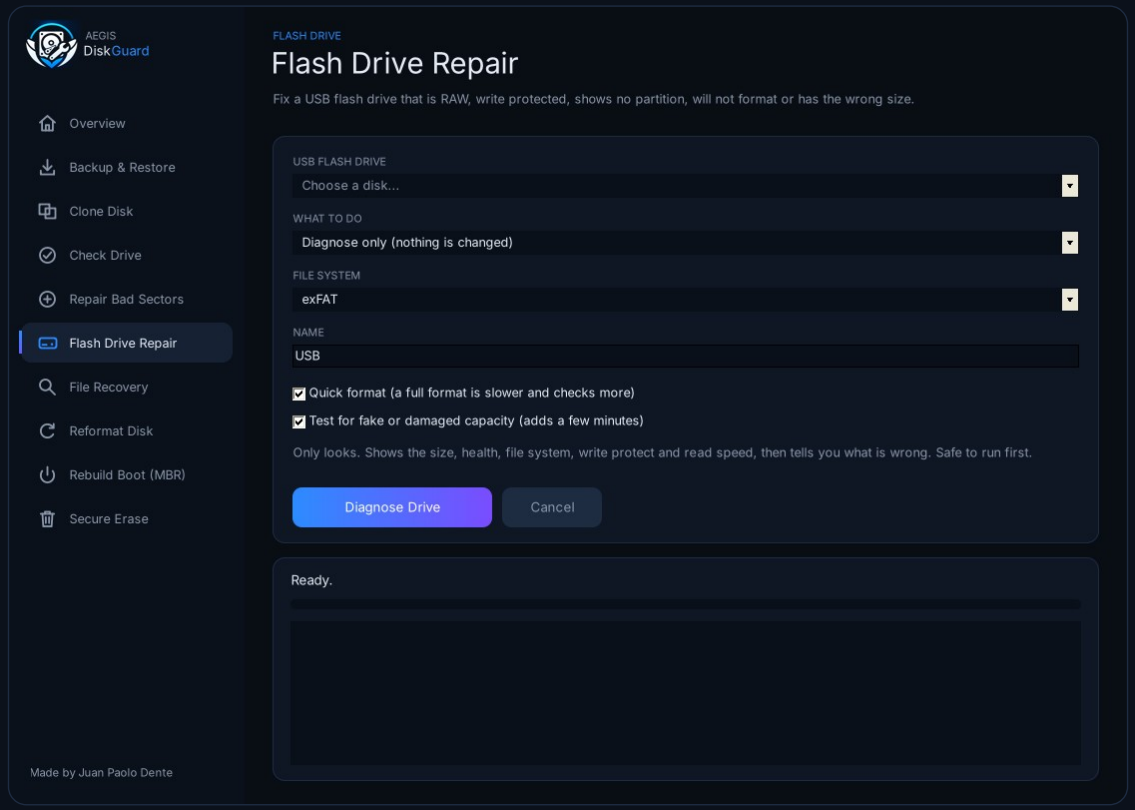
Fixes USB sticks that are RAW, write protected, show no partition or won’t format. Diagnose first, repair second.

Why it matters

Most dead looking sticks have a software problem, and fake capacity drives get caught.

How it works

- Clears write protect, rebuilds the partition
- Writes test patterns across the whole size to spot fakes
- Tells you when the controller is locked



TOOLKIT · FUNCTION 7 OF 13

File Recovery

What it does

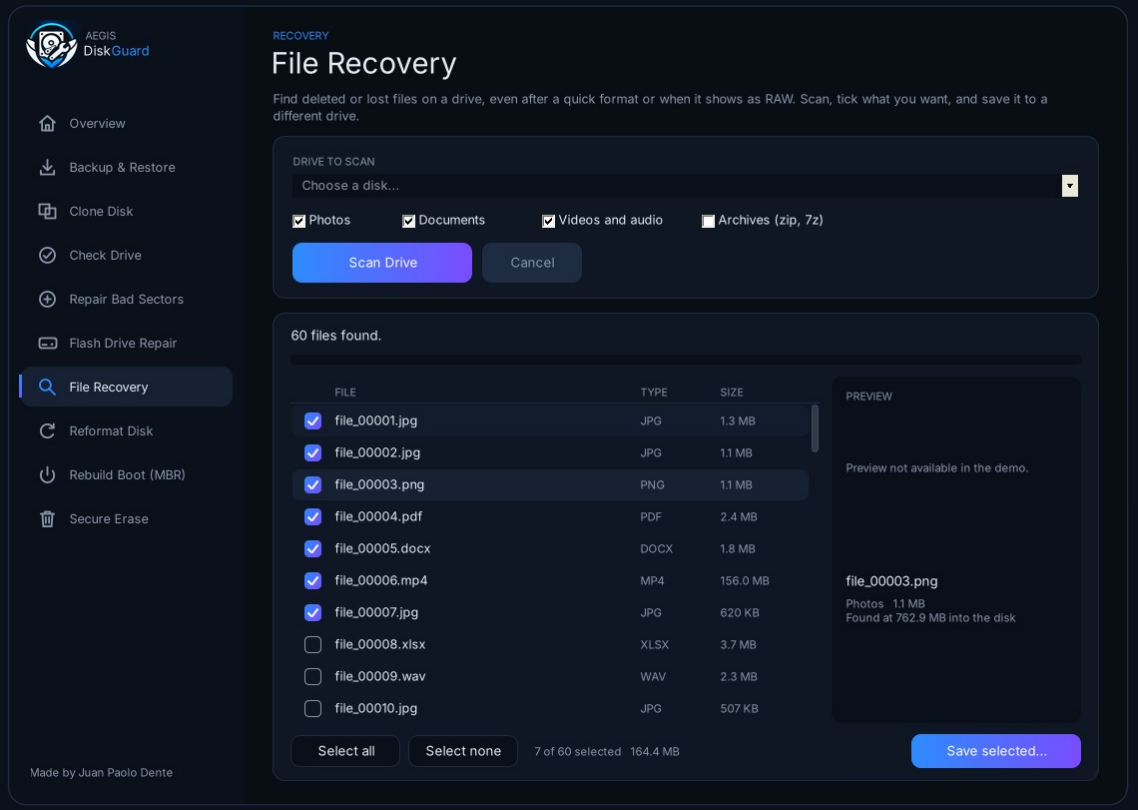
Scans a whole drive for deleted or lost photos, documents, videos and archives, even after a quick format.

Why it matters

Deleting only removes the index entry. The data often stays until something writes over it.

How it works

- Finds file starts by their byte signatures
- Measures each file by reading its structure
- Refuses to save onto the scanned drive



TOOLKIT · FUNCTION 8 OF 13

Reformat Disk

What it does

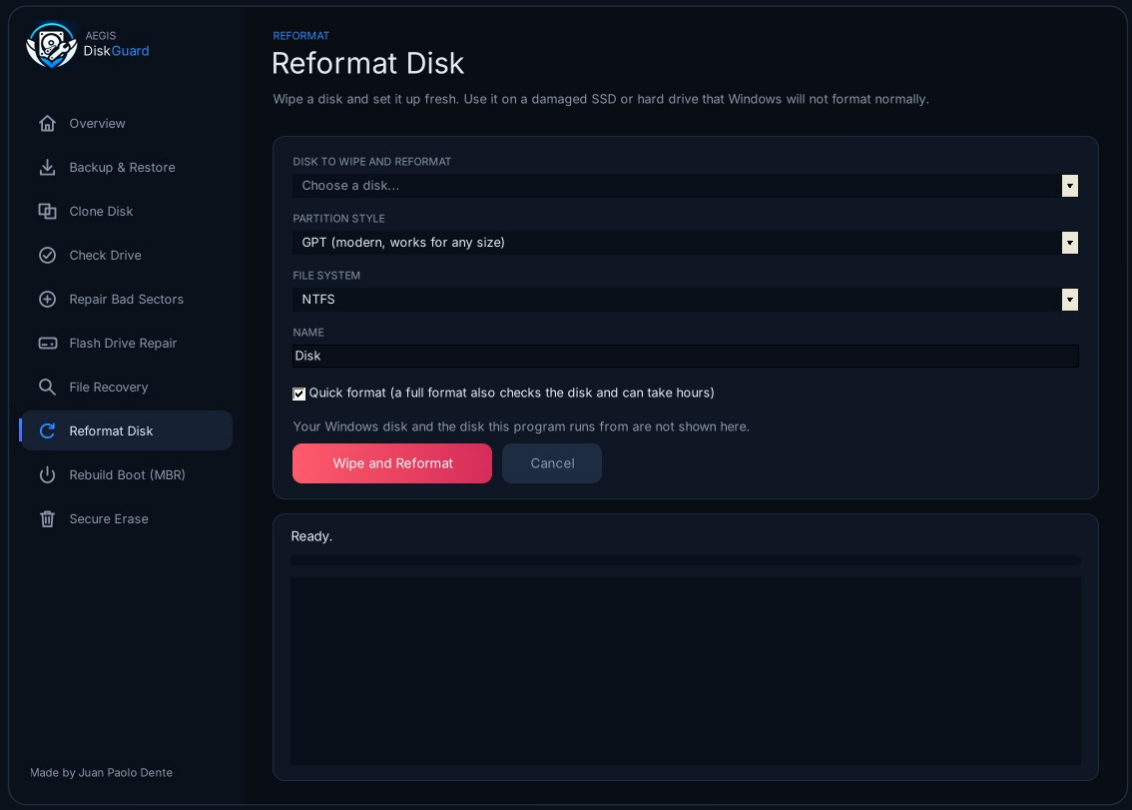
Wipes a disk and sets it up fresh with GPT or MBR, and NTFS, exFAT or FAT32.

Why it matters

Fixes a damaged SSD or hard drive that Windows won’t format the normal way.

How it works

- Clears read only and offline flags first
- Quick or full format
- Hides your Windows disk from the list



TOOLKIT · FUNCTION 9 OF 13

Rebuild Boot (MBR)

What it does

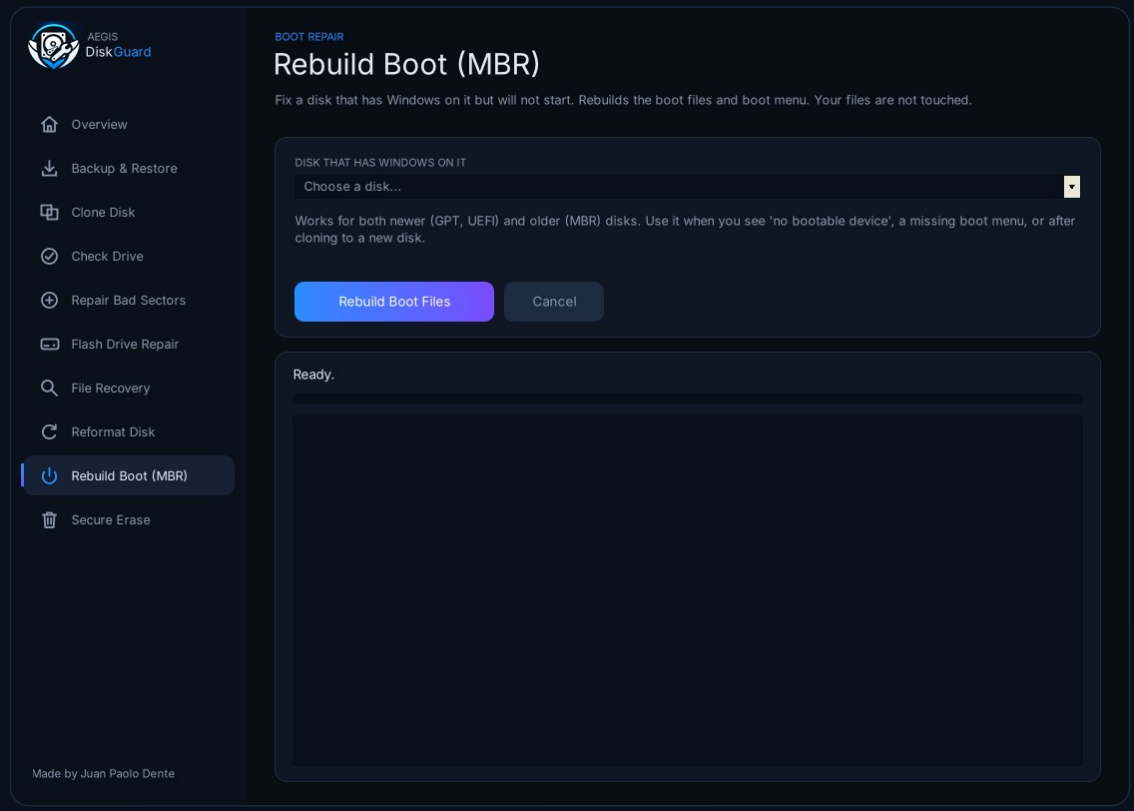
Rebuilds the boot files and boot menu on a disk that has Windows but won’t start. Your files are not touched.

Why it matters

Fixes “no bootable device” and boot problems after cloning to a new disk.

How it works

- Finds the Windows partition by itself
- Runs bcdboot for UEFI or BIOS
- Writes the MBR boot code when needed



TOOLKIT · FUNCTION 10 OF 13

Secure Erase

What it does

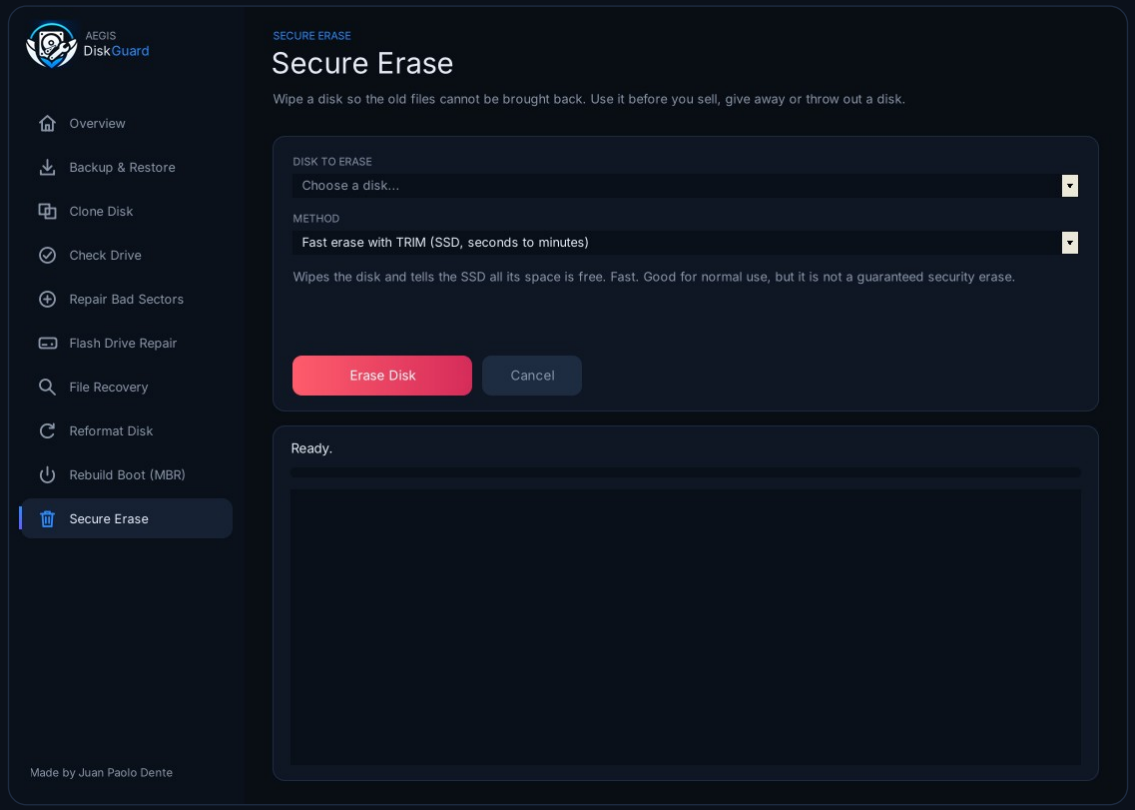
Wipes a disk so old files can’t be brought back: fast TRIM erase, or one pass of zeros or random data.

Why it matters

Use it before you sell, give away or throw out a disk.

Honest limit

This is not the hardware Secure Erase from the SSD maker. The app says so, and points to the maker’s tool for top security.



TOOLKIT · FUNCTION 11 OF 13

Extend Space

What it does

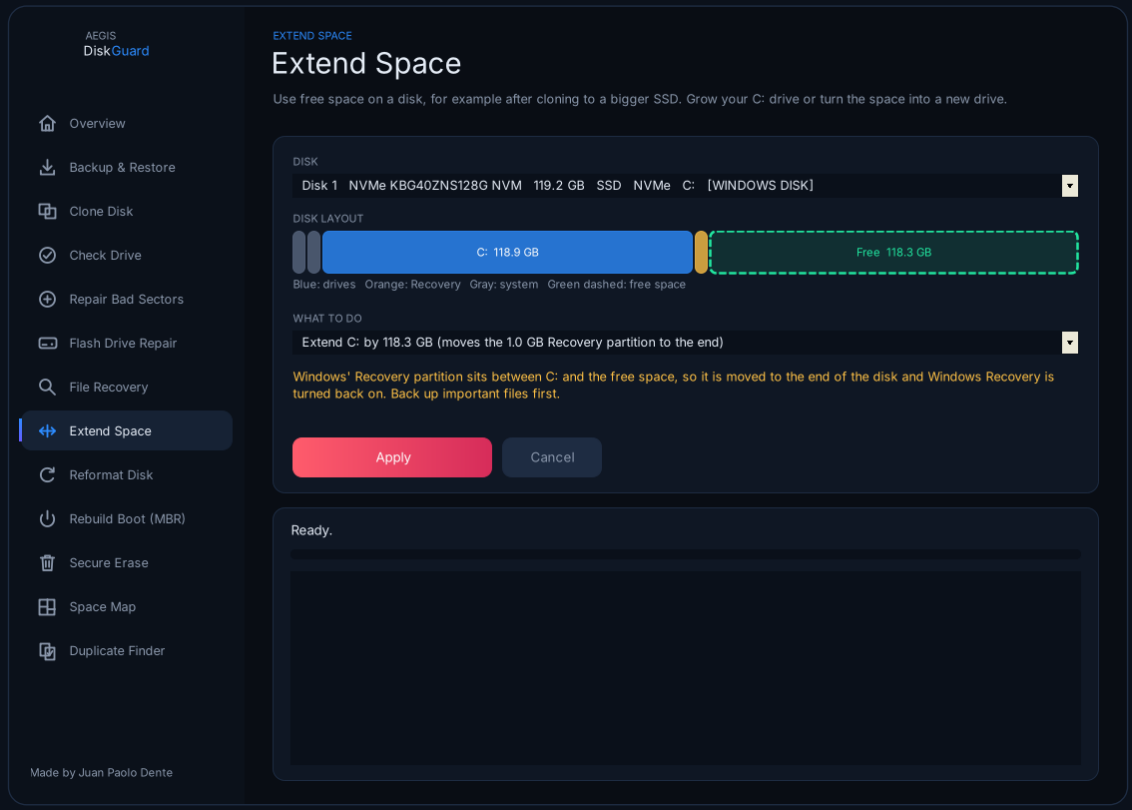
Uses free space on a disk. It can grow C: over it, even when the small Recovery partition is in the way, or turn the space into a new drive.

Why it matters

After cloning to a bigger SSD, Windows still shows the old size. This fixes it without another program.

Honest limit

Moving the Recovery partition changes the disk layout, so it asks you to confirm. It failed once on my own laptop, so I fixed the Windows disk tool step by hand and improved the error message.



TOOLKIT · FUNCTION 12 OF 13

Space Map

What it does

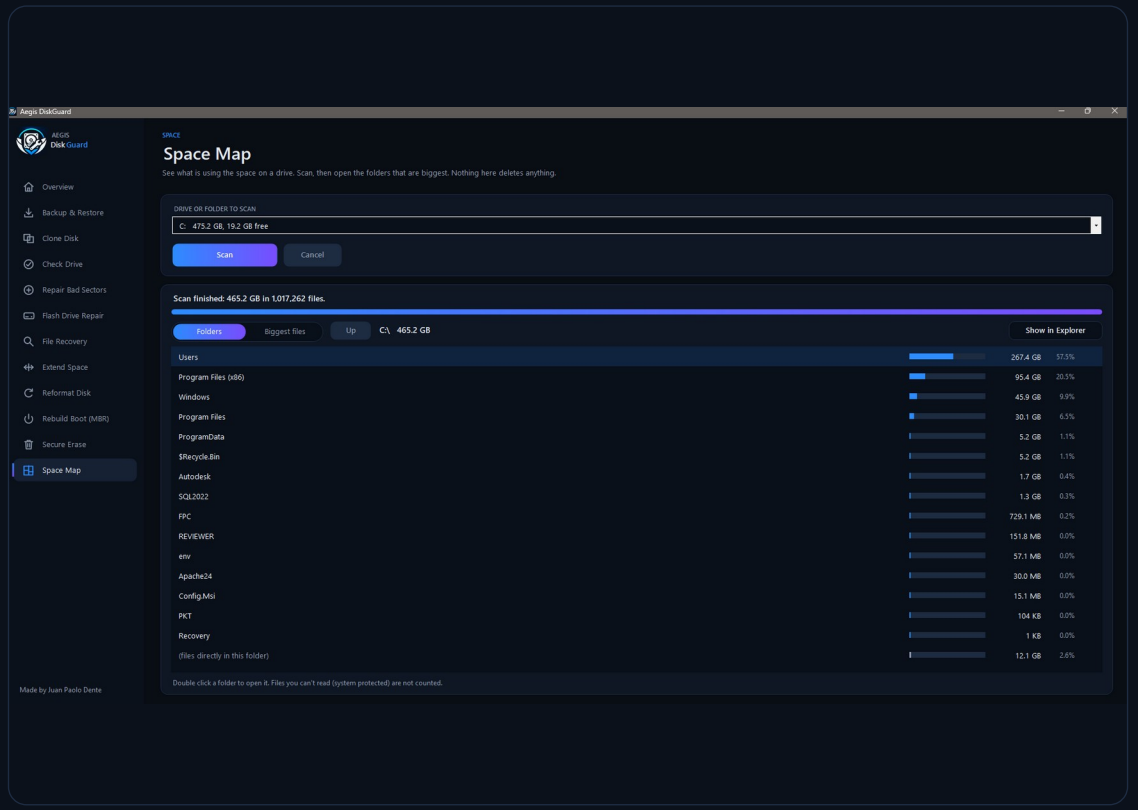
Scans a drive or folder and lists the folders and files that use the most space. Double click a folder to look inside it.

Why it matters

When a disk is almost full, you can see what is eating it in one minute, not by guessing.

Honest limit

It only reads and never deletes. Files that Windows protects can’t be read, so they are not counted.



TOOLKIT · FUNCTION 13 OF 13

Duplicate Finder

What it does

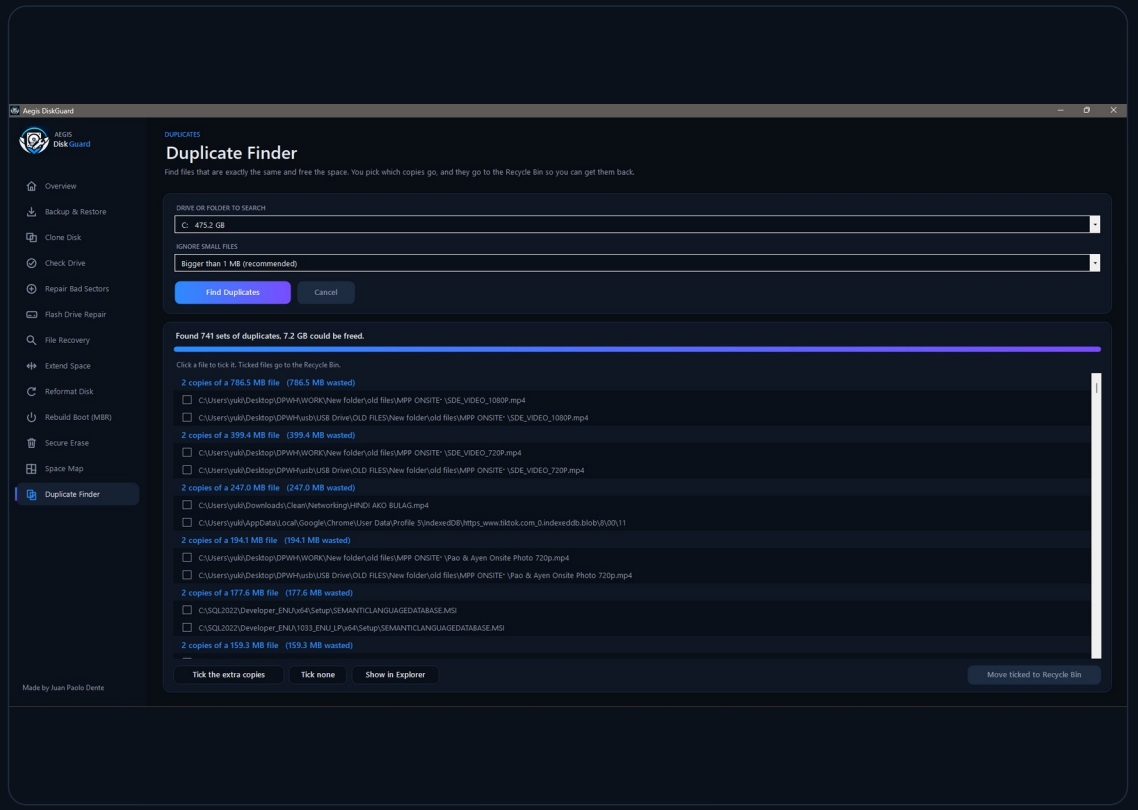
Finds files that are exactly the same, even with different names. You tick the extra copies and they go to the Recycle Bin.

Why it matters

Photos, downloads and backups get copied again and again. Removing the extras frees real space.

Honest limit

It always keeps one copy of every file and skips Windows folders. The Recycle Bin step still needs more tests on real drives.



THE PRINCIPLE

Say what’s real. Never promise what software can’t do.

Marking bad sectors on an SSD does very little, and the app tells you so.

Secure Erase is an overwrite, not the maker’s hardware erase, and it says that too.

A flash drive that fails the capacity test is called fake or dying, and is never formatted.

SAFETY RULES

A disk tool must never break the computer

01

Never touch Windows

Your Windows disk and the disk the app runs from are hidden from every destructive tool.

02

Type to confirm

Before a disk is erased you type its disk number. No accidental clicks.

03

Clones stay offline

A copy shares the original’s disk ID, so Windows is not allowed to bring both online.

04

Save elsewhere

Backups and recovered files can’t be saved onto the disk they came from.

THE REAL TEST

From a stuck 0% to a clone that booted

1 · THE STALL

A clone tool sat at 0% with a 14 hour estimate.

2 · OWN TOOL

Built a raw disk cloner in C# that runs with no install.

3 · THE FAILURE

First run failed near 25% with a data error on the SSD stick.

4 · THE FIX

Heat was the likely cause. Cover off, a fan on, rerun.

5 · THE RESULT

Finished at about 121 MB/s. The check only found live disk changes.

6 · THE BOOT

Booted from the clone, swapped it into the M.2 slot, apps all worked.

ENGINEERING

Small, built in steps, and honest about testing

6.1k

lines of C# in nine source files

13

pages: the dashboard and twelve tools

0

installs needed: it builds with the compiler already in Windows

121

MB/s real clone speed on my own laptop

7

file kinds found with the right size on a test disk image

Not yet

Most tools that change a disk still need more real drive tests

ROADMAP

From one cloner to a full toolkit

STEP 1

Easy Disk Cloner

One window, one job: clone the Windows disk.

STEP 2

The toolkit

Backup, check, repair, reformat, boot and erase in a dark UI.

STEP 3

Brand and dashboard

Aegis DiskGuard logo, clickable disks and stat cards.

STEP 4

Flash Drive Repair

Diagnose, clear write protect, test capacity, reformat.

STEP 5

File Recovery

Signature scan with preview, tested on a disk image.

STEP 6

Disk space tools

Extend Space, Space Map and Duplicate Finder.

NEXT

Real drive tests

NTFS deleted file scan and a Cleaner.

WHAT MAKES IT DIFFERENT

Many disk tools exist. One simple, honest one is rare.

All in one place

Backup, clone, repair, recovery and erase in a single app.

Plain words

Every tool explains what it will do before it does it.

Honest limits

It says “this needs a new drive” instead of pretending to fix it.

Nothing to install

One .exe, built with the compiler Windows already has.

REFLECTION · DRAFT, EDIT IN YOUR OWN WORDS

What I learned, and what’s next

Learned

- How raw disk reads and writes really work in Windows.
- Real failures (a hot SSD stick, a resize that errored) taught more than any test.
- Saying “not yet” builds more trust than a fake success.
- Build for a beginner: plain words beat extra buttons.

Next

- Test every tool on real spare drives.
- NTFS deleted file scan that keeps real file names.
- A Cleaner for temp files, after more real tests.
- Sign the program so Windows stops warning at launch.



AEGIS DiskGuard

Scan. Repair. Protect.

Thank you, questions?

Made by Juan Paolo Dente · [demo video link]